

Deepskilling-DataIntegration

Task 2: NORMALIZATION ANALYSIS

- - 1NF (First Normal Form):
 - All tables contain atomic values.
 - Each column stores a single value only.
- - 2NF (Second Normal Form):
 - All non-key attributes depend on the whole primary key.
 - In the enrollments table, enrollment_date and grade depend on the combination of student_id and course_id, not on either column alone. Therefore there are no partial dependencies.
- - 3NF (Third Normal Form):
 - No non-key attribute depends on another non-key attribute.
 - Department information is stored in the departments table and linked through department_id.
 - Storing dept_name directly in students would create a transitive dependency:
 - Therefore the schema satisfies 3NF.
- - 3NF Analysis for Enrollments Table:
 - The enrollments table contains enrollment_id, student_id, course_id, enrollment_date, and grade.
 - enrollment_date and grade depend directly on the enrollment record. There are no attributes that depend on other non-key attributes.
 - Hence, the enrollments table satisfies 3NF.

Task 3

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Name: Sadhana Shree B

Add a column phone_number VARCHAR(15) to the students table.

```
97
98 • alter table students add column phone_number varchar(15);
99 • alter table courses add column max_seats int default 60;
100 • describe students;
101 • describe courses;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	student_id	int	NO	PRI	NULL	auto_increment
	first_name	varchar(50)	NO		NULL	
	last_name	varchar(50)	NO		NULL	
	email	varchar(100)	NO	UNI	NULL	
	date_of_birth	date	YES		NULL	
	department_id	int	YES	MUL	NULL	
	enrollment_year	year	YES		NULL	
	phone_number	varchar(15)	YES		NULL	

Add a column max_seats INT DEFAULT 60 to the courses table.

cts_sql_exercises SQL File 5*

Limit to 1000 rows

```

93      ('Dr. Meena Pillai', 'meena.p@college.edu', 1, 88000.00),
94      ('Dr. Sunil Rajan', 'sunil.r@college.edu', 2, 82000.00),
95      ('Dr. Latha Gopal', 'latha.g@college.edu', 3, 79000.00),
96      ('Dr. Kartik Bose', 'kartik.b@college.edu', 4, 76000.00);
97
98 •    alter table students add column phone_number varchar(15);
99 •    alter table courses add column max_seats int default 60;
100 •   describe students;
101 •   describe courses;
102      -- Add a CHECK constraint to enrollments ensuring grade is one of ('A','B','C','D'
103 •   alter table enrollments
104      add constraint chk_grades

```

Result Grid

Field	Type	Null	Key	Default	Extra
course_id	int	NO	PRI	NULL	auto_increment
course_name	varchar(100)	NO		NULL	
course_code	varchar(20)	NO	UNI	NULL	
credits	int	NO		NULL	
department_id	int	YES	MUL	NULL	
max_seats	int	YES		60	

Add a CHECK constraint to enrollments ensuring grade is one of ('A','B','C','D','F') or NULL.

```

102      -- Add a CHECK constraint to enrollments ensuring grade is one of ('A','B','C','D','F') or NULL.
103 •   alter table enrollments
104      add constraint chk_grades
105      check (grade in ('A','B','C','D','F') or grade is null);
106 •   describe enrollments;
107      -- Rename the hod name column in departments to head of dept (MySQL: ALTER TABLE ... CHANGE ...).

```

Result Grid

Field	Type	Null	Key	Default	Extra
enrollment_id	int	NO	PRI	NULL	auto_increment
student_id	int	NO	MUL	NULL	
course_id	int	NO	MUL	NULL	
enrollment_date	date	YES		NULL	
grade	char(2)	YES		NULL	

Rename the hod_name column in departments to head_of_dept (PostgreSQL:
ALTER COLUMN ...

```
107      -- Rename the hod_name column in departments to head_of_dept (MySQL: ALTER TABLE ... CHANGE ...).
108 •    alter table departments
109      change hod_name head_of_dept varchar(100);
110 •    describe departments;
111      -- Drop the phone_number column you added in step 1 (simulate a schema rollback).
112 •    alter table students
113      drop column phone_number;
114
```

Field	Type	Null	Key	Default	Extra
department_id	int	NO	PRI	NULL	auto_increment
dept_name	varchar(100)	NO		NULL	
head_of_dept	varchar(100)	YES		NULL	
budget	decimal(12,2)	YES		NULL	

Drop the phone_number column you added in step 1 (simulate a schema rollback)

```
111      -- Drop the phone_number column you added in step 1 (simulate a schema rollback).
112 •    alter table students
113      drop column phone_number;
114 •    describe students;
```

Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	auto_increment
first_name	varchar(50)	NO		NULL	
last_name	varchar(50)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	
date_of_birth	date	YES		NULL	
department_id	int	YES	MUL	NULL	
enrollment_year	year	YES		NULL	